

CLOUD COMPUTING

LAB 01

Ubuntu Server Installation and Configuration

Name: Muskan Zeeshan

Registration No: FA24B1-SE-054

Section: BSSE IV-B

Course Code: CC

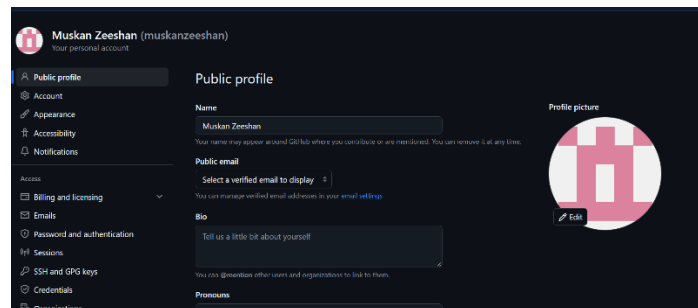
GitHub Username: muskanzeeshan

Task 1: Create and Configure Your GitHub Account

- screenshot clearly showing your profile name and GitHub username

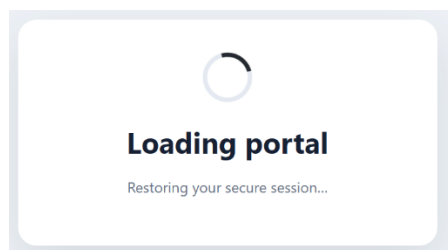


- screenshot showing the completed Name field



Task 2: Register in the Student Marks Portal

- screenshot of the portal login page



- screenshot showing VMware Workstation Pro running

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-54

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section B

Submit enrollment

Sign out

- screenshot showing your GitHub profile information inside the portal

STUDENT MARKS PORTAL

Sign out

Academic Record



muskanzeeshan
GitHub authenticated

Registration
2024-BSE-54

GitHub username
[muskanzeeshan](#)

Email
muskanzeeshan3726@gmail.com

Approved courses
1

+ Enroll in another course

APPROVED COURSES

Course Marks

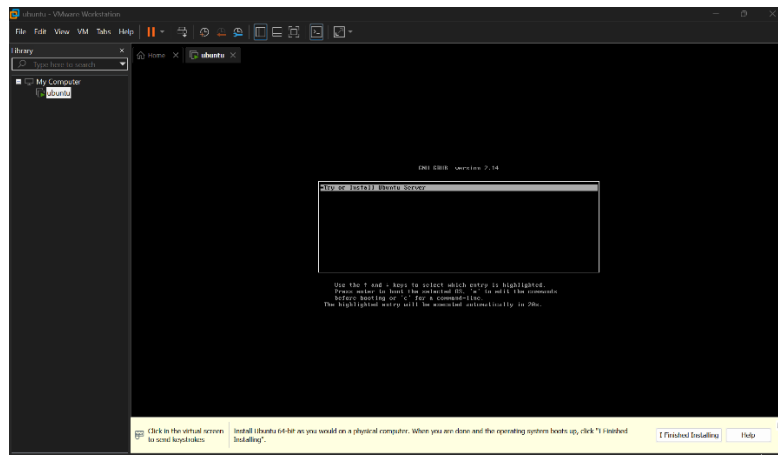
Cloud Computing (SE-665) Section B

Published

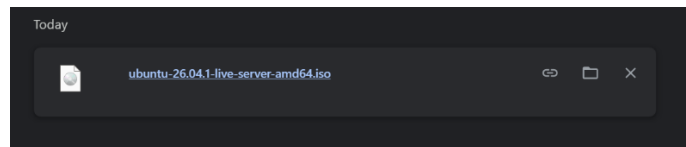
- screenshot of the submitted or pending enrollment

Task 3: Download VMware Workstation Pro and Ubuntu Server

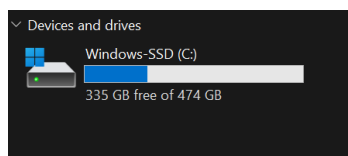
- Download and install VMware Workstation Pro on your Windows computer.



- screenshot showing the downloaded Ubuntu Server ISO filename



- screenshot showing the available storage

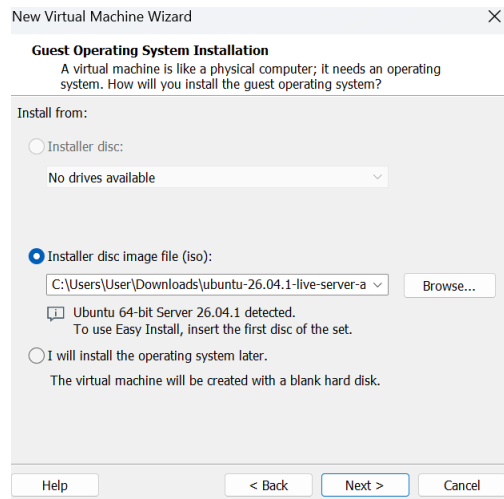


Task 4: Create the Ubuntu Server Virtual Machine

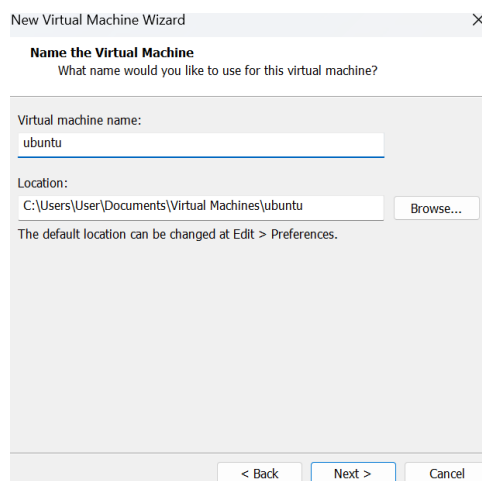
- screenshot of the New Virtual Machine Wizard



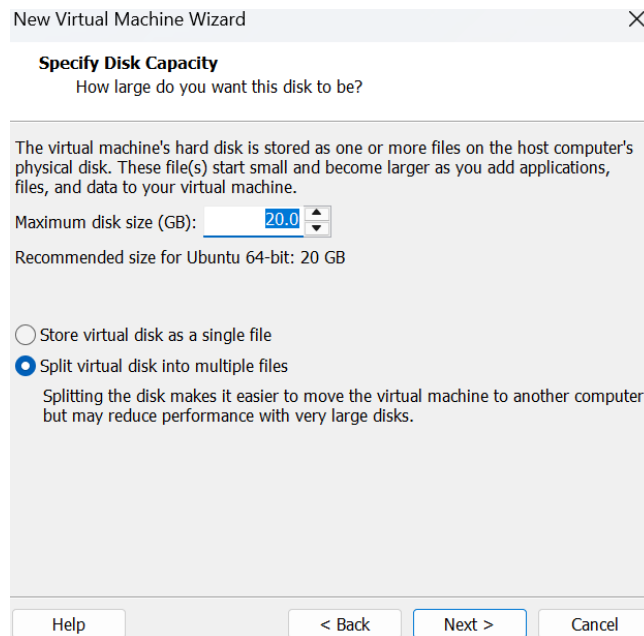
- screenshot showing this selection



- screenshot showing the selected Ubuntu ISO path



- screenshot showing the final virtual machine configuration

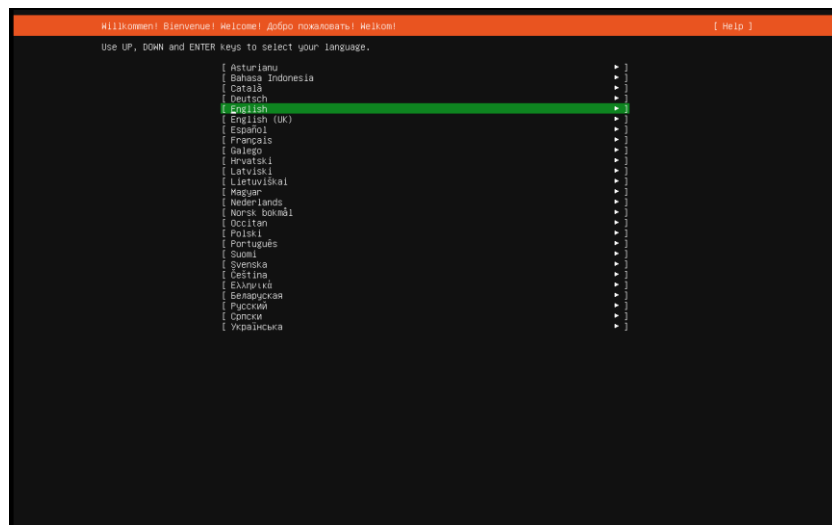


- screenshot of the Ubuntu Server boot screen

```
[ OK ] Listening on systemd-hostnamed.socket - Hostname Service Socket.
[ OK ] Listening on systemd-logind-varlink.socket - User Login Management Varlink Socket.
[ OK ] Listening on uuid.socket - UUID daemon activation socket.
Starting blk-availability.service - Availability of block devices...
[ OK ] Listening on lld-installer.socket - Helper to install lld snap on demand.
[ OK ] Listening on snappy.socket - Socket activation for snappy daemon.
[ OK ] Listening on sshd-vsock.socket - OpenSSH Server Socket (systemd-ssh-generator, AF_VSOCK).
[ OK ] Reached target sockets.target - Socket Units.
[ OK ] Reached target basic.target - Basic System.
[ OK ] Reached target ssh-access.target - SSH Access Available.
Starting chrony.service - chrony, an NTP client/server...
Starting cloud-config.service - Cloud-init: Config Stage...
Starting dbus.service - D-Bus System Message Bus...
[ OK ] Started dmccg.service - Save initial kernel messages after boot.
Starting dracut-shutdown.service - Restore /run/initramfs on shutdown...
Starting e2scrub_reap.service - Remove Stale Online ext4 Metadata Check Snapshots...
[ OK ] Reached target getty.target - Login Prompts.
Starting networkd-dispatcher.service - Dispatcher daemon for systemd-networkd...
Starting nvme-autoconnect.service - Connect NVMe-oF subsystems automatically during boot...
Starting polkit.service - Authorization Manager...
Starting syslog.service - System Logging Service...
[ OK ] Reached target getty-pre.target - Preparation for Logins.
Starting snapd.seeded.service - Wait until snapd is fully seeded...
Starting snapd.service - Snap Daemon...
Starting ssh.service - OpenSSH Secure Shell server...
Starting sysstat.service - Resets System Activity Logs...
Starting systemd-logind.service - User Login Management...
Starting udisks2.service - Disk Manager...
[ OK ] Finished blk-availability.service - Availability of block devices.
[ OK ] Finished dracut-shutdown.service - Restore /run/initramfs on shutdown.
[ OK ] Started ssh.service - OpenSSH Secure Shell server.
[ OK ] Finished nvme-autoconnect.service - Connect NVMe-oF subsystems automatically during boot.
[ OK ] Finished sysstat.service - Resets System Activity Logs.
[ OK ] Reached target remote-fs-pre.target - Preparation for Remote File Systems.
[ OK ] Reached target remote-fs.target - Remote File Systems.
Starting apport.service - automatic crash report generation...
[ OK ] Started cron.service - Regular background program processing daemon.
Starting multipathd-queueing.service - Enable queueing for multipath maps...
Starting systemd-user-sessions.service - Permit User Sessions...
[ OK ] Finished multipathd-queueing.service - Enable queueing for multipath maps.
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
Starting plymouth-quit-wait.service - Hold until boot process finishes up...
Starting plymouth-quit.service - Terminate Plymouth Boot Screen...
[ OK ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
Starting setvtrgb.service - Set console scheme...
[ OK ] Finished plymouth-quit.service - Terminate Plymouth Boot Screen.
[ OK ] Started systemd-logind.service - User Login Management.
[ OK ] Started unattended-upgrades.service - Unattended Upgrades Shutdown.
[ OK ] Finished e2scrub_reap.service - Remove Stale Online ext4 Metadata Check Snapshots.
-
```

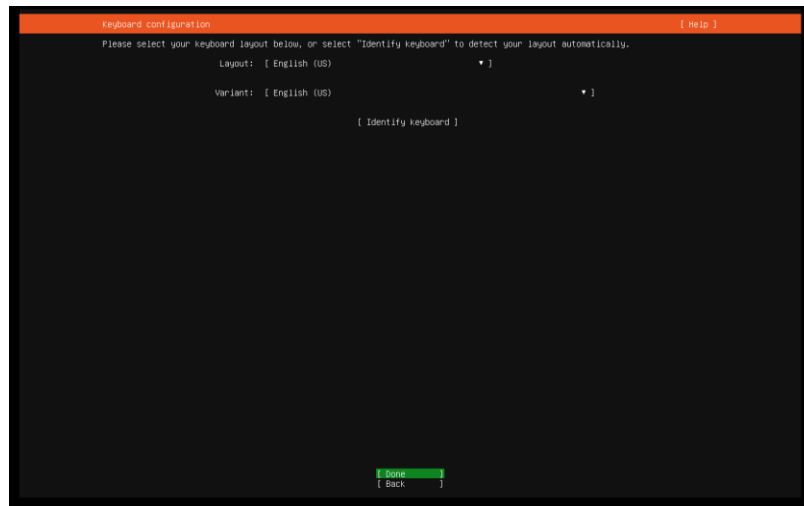
Task 5: Choose the Installation Language

- screenshot showing the selected language

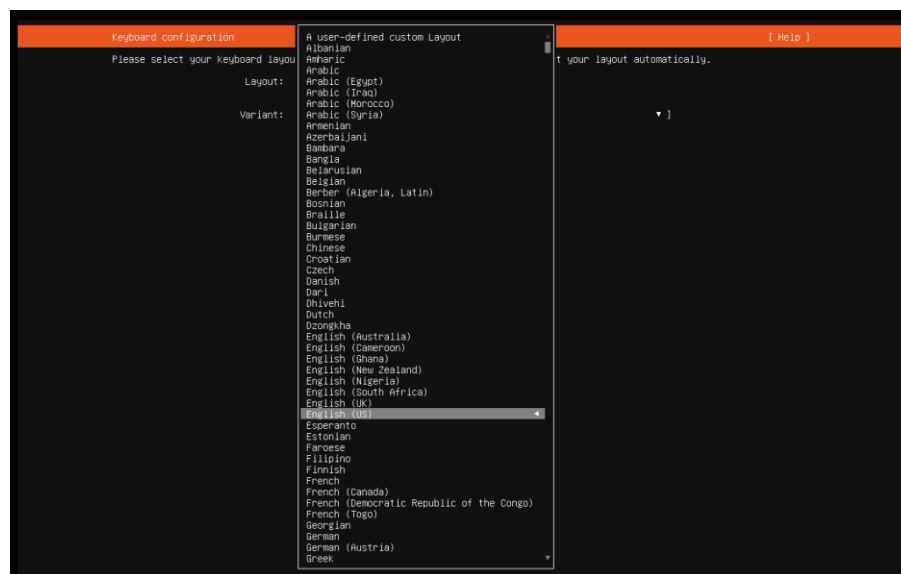


Task 6: Configure the Keyboard Layout

- screenshot showing the selected keyboard layout

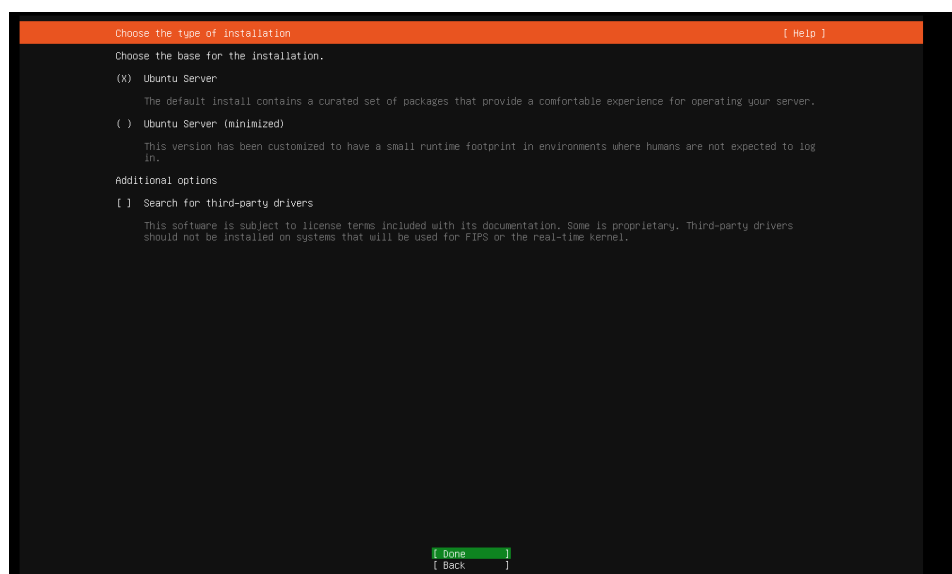


- screenshot showing the selected keyboard variant



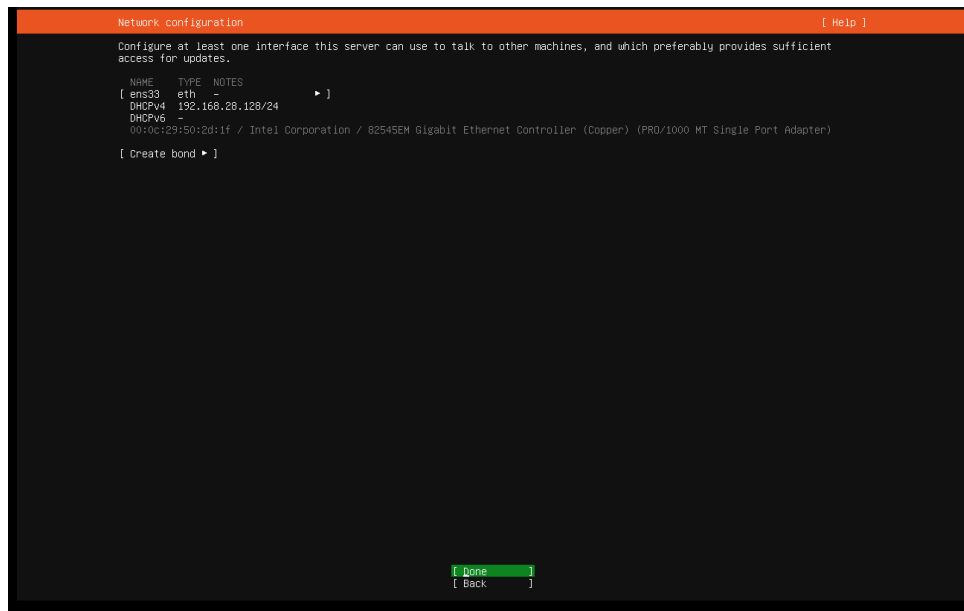
Task 7: Choose the Ubuntu Server Installation Type

- screenshot showing the selected installation type

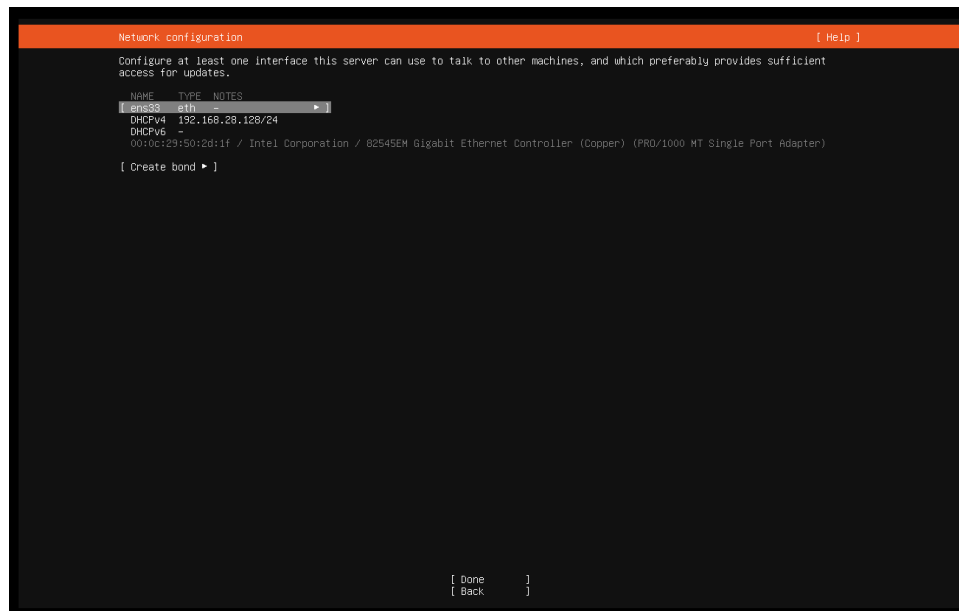


Task 8: Configure the Network

- screenshot showing the detected network interface

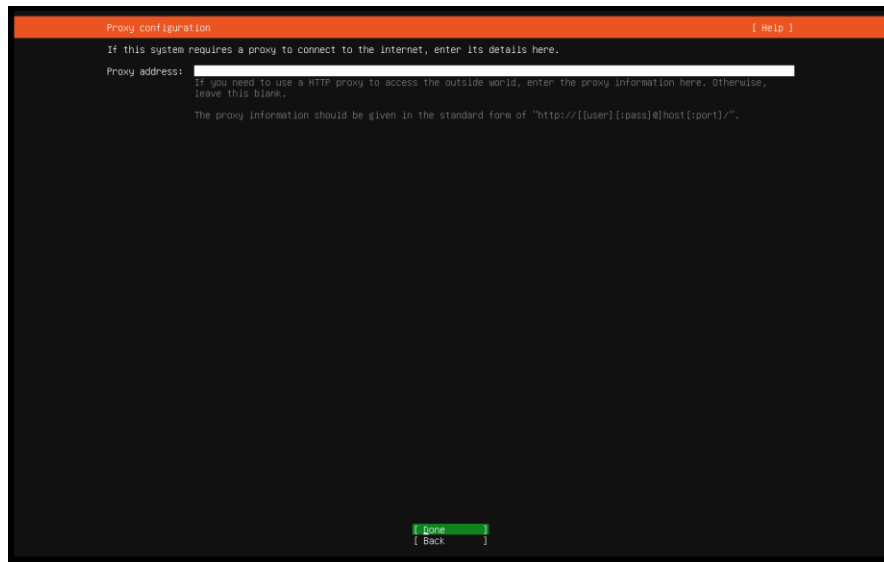


- screenshot showing the assigned network address

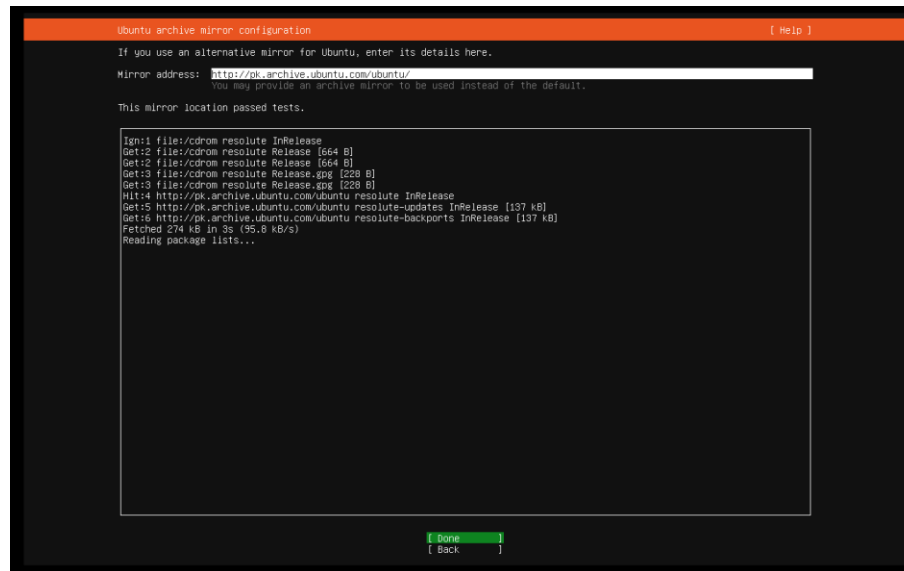


Task 9: Configure Storage

- screenshot showing the selected guided storage option

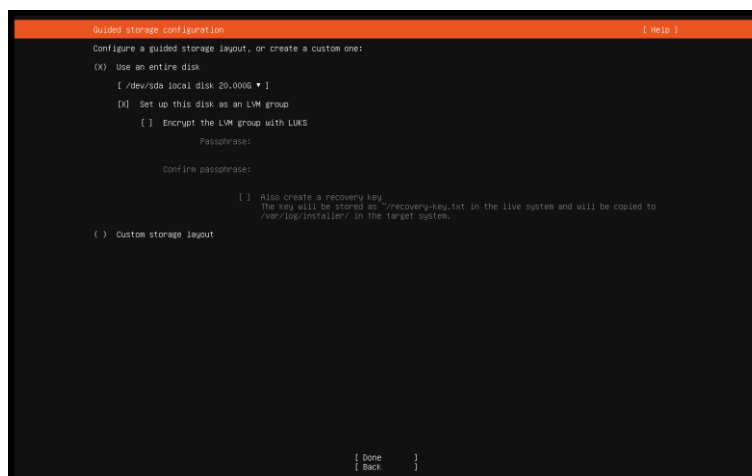


- screenshot of the storage configuration

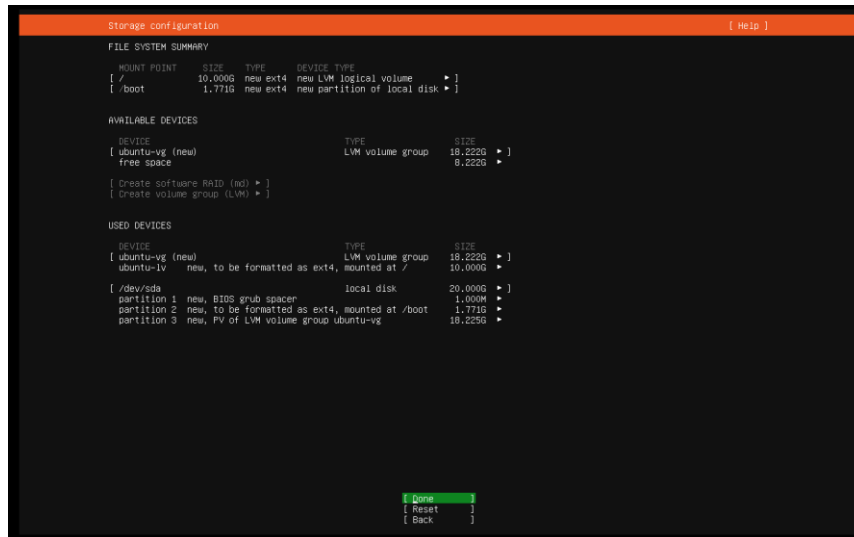


Task 10: Select the Installation Disk

- screenshot showing the selected disk

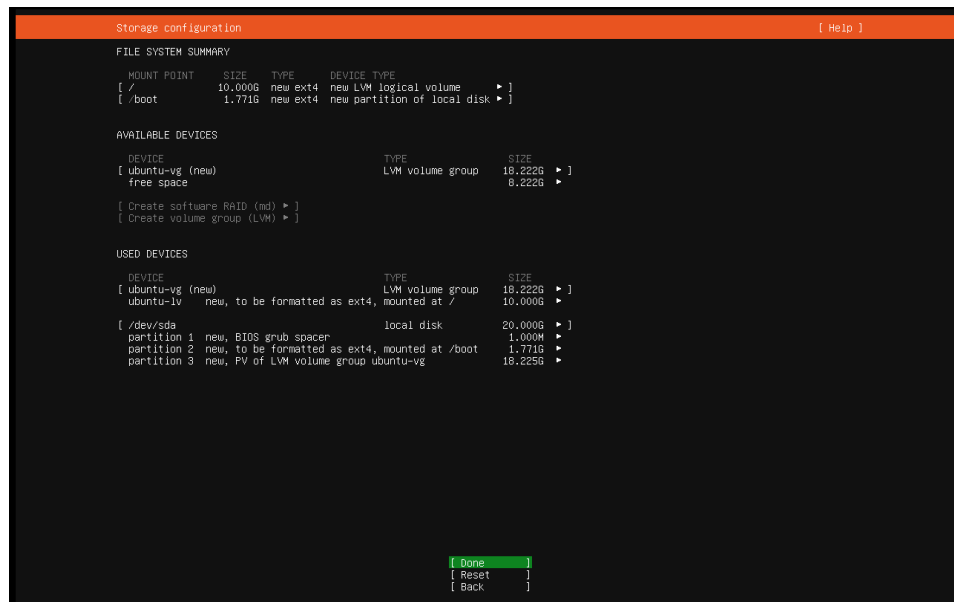


- screenshot showing the disk name and size



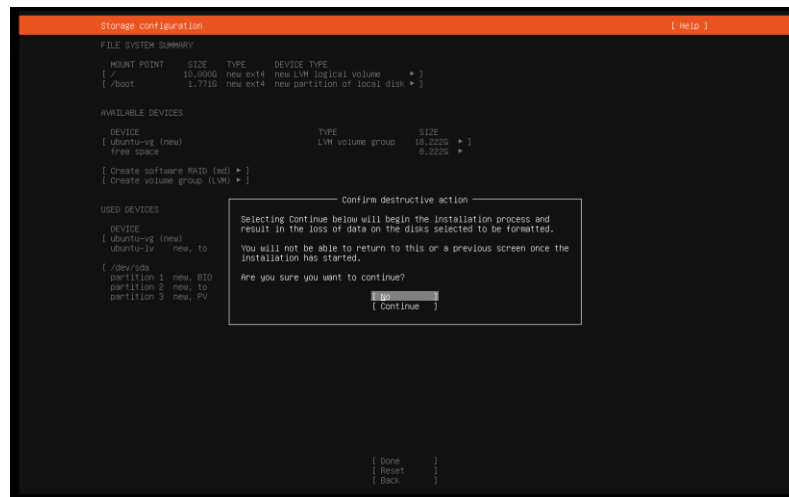
Task 11: Review the Partitions

- screenshot showing the proposed partitions

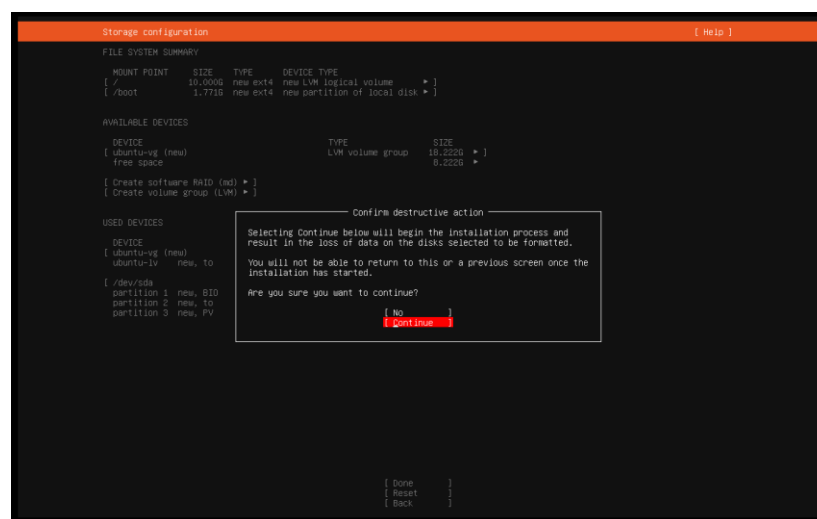


Task 12: Confirm the Storage Changes

- screenshot showing the final warning

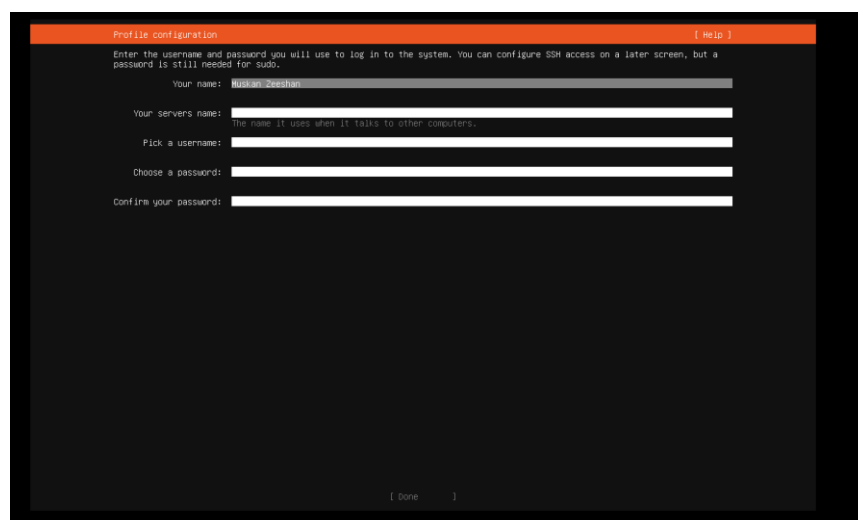


- screenshot showing the confirmation selection



Task 13: Configure the Ubuntu Profile

- screenshot showing the entered name



- screenshot showing the server name

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Muskan Zeeshan

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

➤ screenshot showing the Ubuntu username

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Muskan Zeeshan

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: muskanzeeshan

Choose a password:

Confirm your password:

[Done]

➤ screenshot of the completed profile page

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Muskan Zeeshan

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: muskanzeeshan

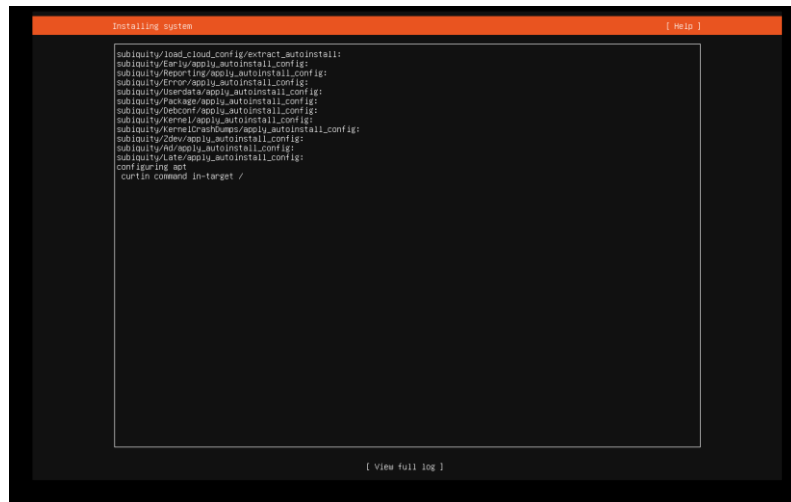
Choose a password:

Confirm your password:

[Done]

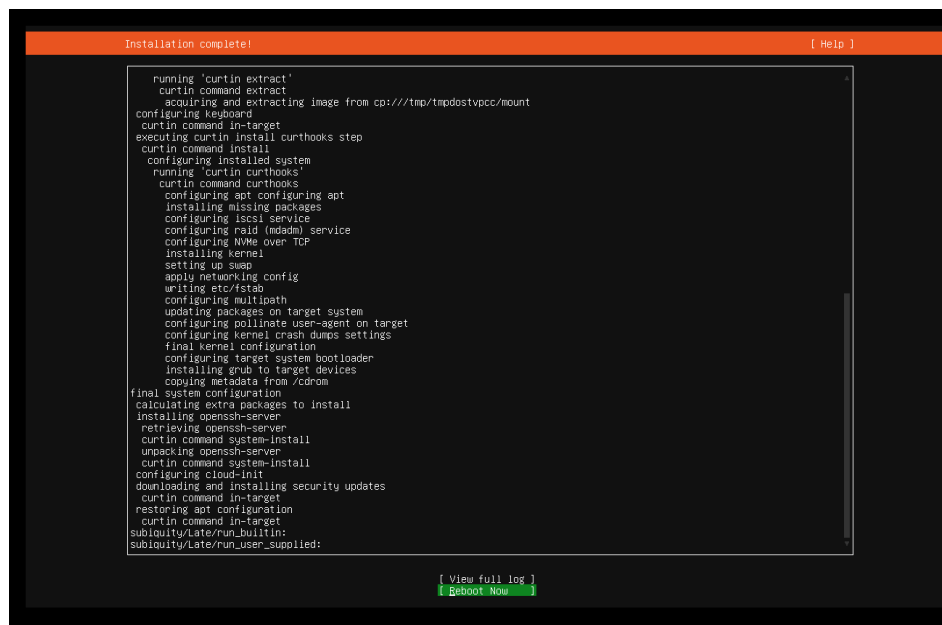
Task 14: Complete the Software Installation

- screenshot showing the installation progress

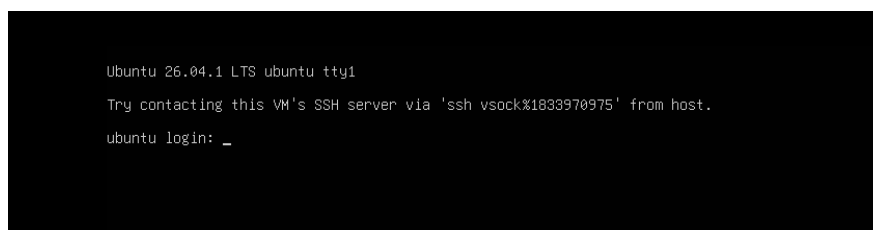


Task 15: Reboot and Verify the Ubuntu Identity

- screenshot of the installation-complete screen



- screenshot showing the Ubuntu login screen



- screenshot showing the successful terminal login

```

Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%1833970975' from host.

ubuntu login: muskanzeeshan
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 16:55:00 UTC 2026

System load:  1.1          Processes:      250
Usage of /:   50.7% of 9.75GB Users logged in:  0
Memory usage: 21%         IPv4 address for ens33: 192.168.28.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

muskanzeeshan@ubuntu:~$

```

- screenshot showing the complete prompt

```

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

muskanzeeshan@ubuntu:~$ _

```

Task 16: Find the Ubuntu Server IP Address

- screenshot showing the command and its output

```

muskanzeeshan@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP gro
    link/ether 00:0c:29:50:2d:1f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.28.131/24 metric 100 brd 192.168.28.255 scope global dynamic ens3
        valid_lft 1607sec preferred_lft 1607sec
    inet6 fe80::20c:29ff:fe50:2d1f/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

- screenshot clearly showing the correct IPv4 address

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:50:2d:1f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.28.131/24 metric 100 brd 192.168.28.255 scope global dynamic ens33
        valid_lft 1607sec preferred_lft 1607sec
    inet6 fe80::20c:29ff:fe50:2d1f/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

Task 17: Connect to Ubuntu Server from Windows

- screenshot showing the SSH command

```
Command Prompt - ssh musl X + v
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh muskanzeeshan@192.168.28.131
The authenticity of host '192.168.28.131 (192.168.28.131)' can't be established.
ED25519 key fingerprint is SHA256:7HNUKR+MBx5/n9DRTFypdBwjHd9GWJN+3roMJ1mTX2E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

- screenshot showing the accepted fingerprint

```
Command Prompt - ssh musl X + v
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh muskanzeeshan@192.168.28.131
The authenticity of host '192.168.28.131 (192.168.28.131)' can't be established.
ED25519 key fingerprint is SHA256:7HNUKR+MBx5/n9DRTFypdBwjHd9GWJN+3roMJ1mTX2E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes|
```

- screenshot showing the successful SSH session

```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh muskanzeeshan@192.168.28.131
The authenticity of host '192.168.28.131 (192.168.28.131)' can't be established.
ED25519 key fingerprint is SHA256:7HNUKR+MBx5/n9DRTFypdBwjHd9GWJN+3roMJ1mTX2E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.28.131' (ED25519) to the list of known hosts.
muskanzeeshan@192.168.28.131's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 17:44:19 UTC 2026

System load:  0.01          Processes:      222
Usage of /:   51.2% of 9.75GB Users logged in:  0
Memory usage: 23%          IPv4 address for ens33: 192.168.28.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

muskanzeeshan@ubuntu:~$ |
```

- screenshot showing the remote prompt

```
System information as of Wed Sep 16 17:44:1

System load:  0.01          Processes:
Usage of /:   51.2% of 9.75GB Users logg
Memory usage: 23%          IPv4 addre
Swap usage:   0%

Expanded Security Maintenance for Applicatio

12 updates can be applied immediately.
To see these additional updates run: apt lis

Enable ESM Apps to receive additional future
See https://ubuntu.com/esm or run: sudo pro

muskanzeeshan@ubuntu:~$ |
```